AIRSTREET · DESIGN DECISION RECORD

AirStreet decisions of 2026-10-30

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DOCUMENT CONTROL

Document ID	AST-DDR-006	Version	0.1
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
Source	docs/decisions/0006-decisions-2026-10-30.md @ c748953		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	Records open decisions 1 to 5 of AST-DEC-001 as decided by Amish on 2026-10-30, with what was carried through (tube hood in the model, BOM line 12, AST-DWG-110, PR-004 and PR-019, mass, area and cost)

0006: Decisions of 2026-10-30

- **Date:** 2026-10-30
- **Status:** accepted

Context

After a report listing the proposed calls in four repos, Amish wrote on 2026-10-30: "on the calls - i go with your recommendations". The five open decisions of AST-DEC-001 each carried a stated recommendation, so each is recorded below as decided, with the option recommended.

What was decided

#	Decision	Option taken (the recommendation)	Carried through
1	Concept selection	A, the current design (221 of 260; C 198, B 188; no option within 10 %)	`concept_chosen: A`; AST-DDR-004 accepted; no change to the working principle
2	Diffusion tube cross-check	(a) triplicate NO2 diffusion tubes at the anchor node and one node in five: a printed hood and three tube clips on the cross arm; tubes changed every two to four weeks by the partner; about £8 per tube with analysis (2015 price)	Model (`hood`, `hood_clips`, `hood_screws`), BOM line 12 and line 11, AST-DWG-110, build plan step 17, PR-019, risks H9 and F11
3	Validity above the rated range	(a) now, accept flagged hours (AST-DDR-005 F2); (b) seek a NO2 cell rated above 40 degC with the partner when the partner is agreed	DEC action 7; no design change
4	PR-004 target	(b) classify against both 40 and 20 ug/m3 (the same uncertainty gives 33.9 to 46.1 and 13.9 to 26.1 ug/m3)	PR-004 text and status, `sizing.py` [C9], AST-CAL-001 v0.8, AST-REG-001 A2
5	Text of PR-017 and PR-018	(a) accept as written	PRS and RTM no longer say "proposed"

What changed in the design (decision 2)

The hood hangs under the cross arm between the pod and the shield, on two M5 screws into heat-set inserts, like the pod and the shield cap. The arm keeps its 415 mm length; two more 5.5 mm holes are drilled in its flat leg. Tubes are not part of the node (a laboratory supplies them with analysis); the hood has three sockets and each tube is held by a printed clip.

Quantity	Before	After
Parts in the D3 model of option A	195 (103 fasteners)	205 (109 fasteners); mass in `cad/fidelity.json` 3.4 kg
Mass on the pole	3.61 kg (3.77 kg with the sun shield)	3.70 kg (3.86 kg); margin to 4.0 kg 0.14 kg
Frontal area	0.139 m2	0.144 m2 (limit 0.15 m2)
Wind at 35 m/s	125 N, 416 N m	129 N, 431 N m
Cross arm stress	4.4 MPa	5.1 MPa
Sensor head cost	USD 302.50	USD 306.00 (line 12 USD 2.50 and line 11 USD 1.00, both estimates)
Whole node with FieldNode	USD 433.50	USD 437.00

`budget_usd` stays at 285. PR-015 is over the value-engineering target by USD 21.00 (it was USD 17.50).

Consequences

- The tube change every two to four weeks is a new reason to climb to the pole (hazard H9). It is the partner's visit plan to settle (DEC action 8); the hood is fitted at only the anchor node and one node in five.
- Frontal area has 0.006 m2 of margin and mass 0.14 kg; any further addition to the node needs a trade.
- Decision 3 (b) and the partner agreement for the tubes are actions for Amish, not open decisions.

References

AST-DEC-001, AST-DDR-004, AST-DDR-005, AST-OPT-001, AST-PRS-001, AST-CAL-001, AST-RSK-001.